CISCO CCNA PORTFOLIO PROJECT 2

Department VLAN Segmentation and Inter-VLAN Routing

A hands-on Cisco Packet Tracer implementation

PROJECT HIGHLIGHTS

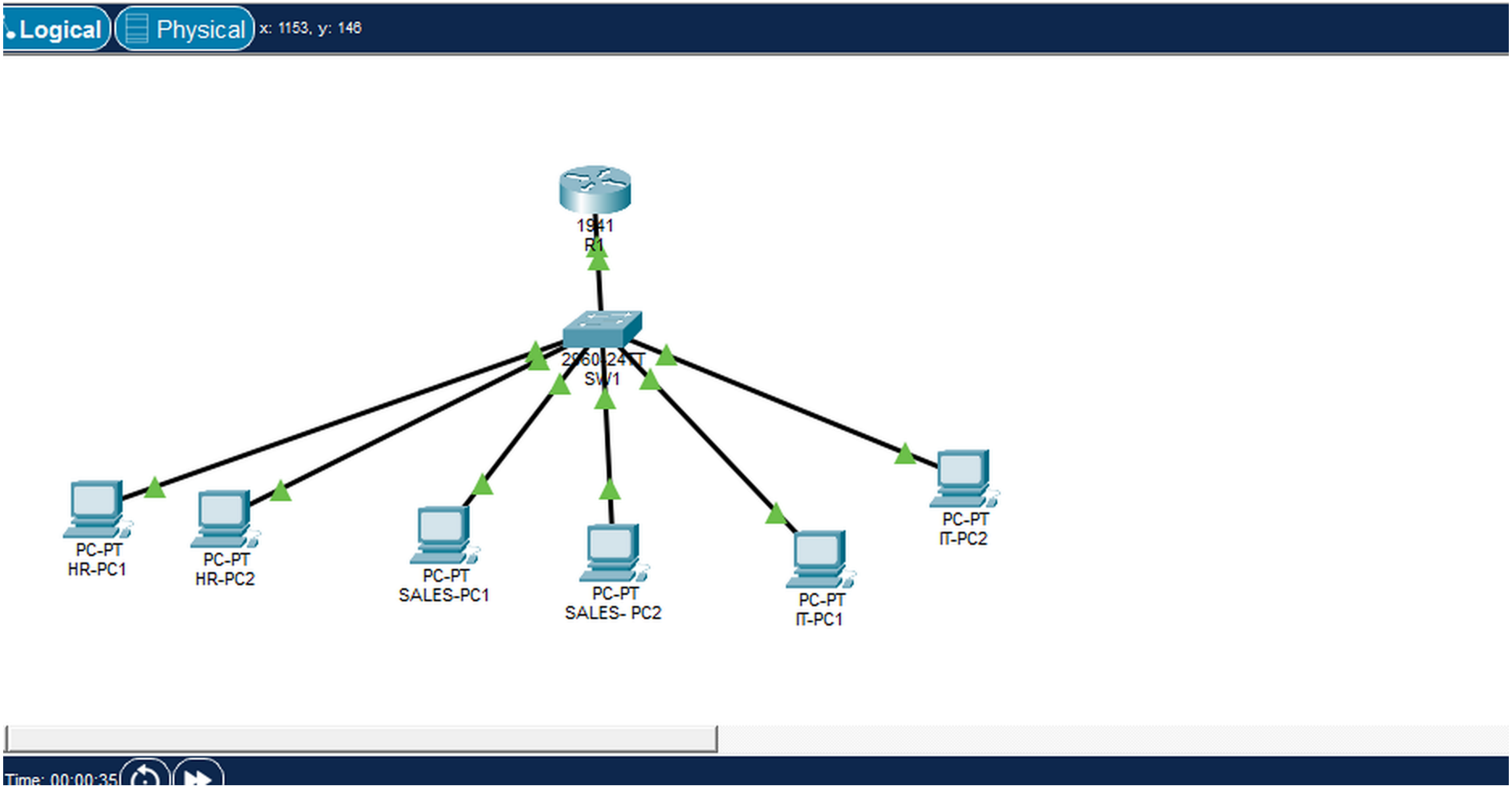
- Three department VLANs: HR, Sales, and IT
- IEEE 802.1Q trunking and router-on-a-stick
- Department-specific DHCP address pools
- Inter-VLAN communication and troubleshooting

LaKayla Stewart

Cisco Packet Tracer | Networking Portfolio | July 2026

Department VLAN Network Topology

VERIFIED



PROJECT EVIDENCE

Designed a departmental network using one Cisco router, one Cisco 2960 switch, and six workstations representing HR, Sales, and IT.

WHAT THIS DEMONSTRATES

- Logical network design
- Clear department labeling
- Router and switch connectivity

VLAN Creation for HR, Sales, and IT

VERIFIED

VLAN 10 - HR

```
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#vlan 10
SW1(config-vlan)#name HR
SW1(config-vlan)#exit
```

VLAN 20 - SALES

```
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#vlan 20
SW1(config-vlan)#name SALES
SW1(config-vlan)#exit
SW1(config)#end
SW1#
```

VLAN 30 - IT

```
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#vlan 30
SW1(config-vlan)#name IT
SW1(config-vlan)#exit
```

Department segmentation created on SW1

PROJECT EVIDENCE

Created VLAN 10 for HR, VLAN 20 for Sales, and VLAN 30 for IT to separate departmental broadcast domains.

WHAT THIS DEMONSTRATES

- VLAN creation and naming
- Department segmentation
- Cisco IOS configuration

ROUTING

Router-on-a-Stick Configuration

VERIFIED

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
```

PROJECT EVIDENCE

Configured IEEE 802.1Q router subinterfaces for VLANs 10, 20, and 30, with each subinterface serving as the department default gateway.

WHAT THIS DEMONSTRATES

- 802.1Q encapsulation
- Inter-VLAN routing
- Default gateway design

IEEE 802.1Q Trunk Verification

VERIFIED

```
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,30

SW1#
```

PROJECT EVIDENCE

Verified that the switch-to-router link was operating as an 802.1Q trunk and carrying VLANs 10, 20, and 30.

WHAT THIS DEMONSTRATES

- Trunk status verification
- Allowed VLAN review
- Layer 2 transport

Department DHCP Lease Verification

VERIFIED

```
R1#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.11    0090.2BEE.5C42           --                     Automatic
192.168.10.12    00D0.BAC5.8C52           --                     Automatic
192.168.20.11    00E0.A305.72E9           --                     Automatic
192.168.20.12    0090.2B32.BBDD           --                     Automatic
192.168.30.11    0030.A32A.6005           --                     Automatic
192.168.30.12    0001.6355.0848           --                     Automatic
R1#
```

PROJECT EVIDENCE

Confirmed six DHCP leases: two addresses each for HR, Sales, and IT, with every workstation receiving an address from the correct network.

WHAT THIS DEMONSTRATES

- DHCP pool operation
- Correct subnet allocation
- Client lease verification

TROUBLESHOOTING

Incorrect VLAN Assignment Identified

ISSUE FOUND

```
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fastethernet 0/4
SW1(config-if)#switchport access vlan 10
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   HR                               active    Fa0/1, Fa0/2, Fa0/4
20   SALES                            active    Fa0/3
30   IT                               active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#
```

PROJECT EVIDENCE

Introduced a controlled fault by placing Fa0/4 in VLAN 10 instead of VLAN 20, causing a Sales workstation to join the HR network.

WHAT THIS DEMONSTRATES

- Fault isolation
- show vlan brief analysis
- Access-port troubleshooting

VALIDATION

Connectivity Restored Across VLANs

VERIFIED

```
Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PROJECT EVIDENCE

Corrected the Sales access-port assignment and verified successful connectivity to the HR, Sales, and IT gateway addresses with zero packet loss.

WHAT THIS DEMONSTRATES

- Configuration correction
- End-to-end ping testing
- Inter-VLAN validation

Technical Outcomes and Skills

VLAN AND ADDRESSING PLAN

VLAN	Department	Network	Gateway
10	HR	192.168.10.0/24	192.168.10.1
20	Sales	192.168.20.0/24	192.168.20.1
30	IT	192.168.30.0/24	192.168.30.1

SKILLS DEMONSTRATED

- Cisco Packet Tracer
- Cisco IOS
- VLAN segmentation
- IEEE 802.1Q trunking
- Router-on-a-stick
- DHCP
- IPv4 addressing
- Network troubleshooting

PROJECT OUTCOME

This project demonstrates the ability to design and configure a segmented departmental network, provide automatic IP addressing, route traffic between VLANs, verify trunk and DHCP operation, and troubleshoot an incorrect access-port assignment. The completed lab provides hands-on evidence of foundational CCNA switching, routing, and troubleshooting skills.

PORTFOLIO-READY DELIVERABLE

Built, verified, troubleshot, documented, and prepared for LinkedIn project media.